INCH-POUND

MIL-S-11716D
12 January 1993
SUPERSEDING
MIL-S-11716C
3 April 1972

MILITARY SPECIFICATION

SUPERSTRUCTURE, STEEL, FOR BRIDGE, FLOATING, PNEUMATIC FLOAT, CLASS 60

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the deck panels, filler panels, curbs, ramp panels, and miscellaneous members comprising the superstructure supported on saddled pneumatic floats of a military-type floating bridge.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: SATBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

FEDERAL

FED-STD-595	- Colors Used in Government Procurement.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

D2777-2 through -7	- Superstructure, Steel.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-JBS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of

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documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B18.2.1	- Square and Hex Bolts and Screws Inch Series Including Hex Cap Screws and Lag Screws.
B18.2.2	- Square and Hex Nuts (Inch Series).
B18.22.1	- Plain Washers.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 36	- Structural Steel.
A 53	- Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless.
A 304	- Alloy Steel Bars Subject to End-Quench Hardenability Requirements.
A 513	- Electric-Resistance-Welded Carbon and Alloy Steel Mechanical Tubing.
A 572/A 572M	- High-Strength Low-Alloy Columbium-Vanadium Steels of Structural Quality.
D 3951	- Packaging, Commercial.
D 3953	- Flat Steel Strap and Connectors.
D 4675	- Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

A5.1	- Covered Carbon Steel Arc Welding Electrodes.
A5.18	- Carbon Steel Filler Metals for Gas Shielded Arc Welding.
D1.1	- Structural Welding Code - Steel.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC. AGENT

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The steel superstructure shall be as shown on drawings D2777-2 through -7 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the superstructure. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. The materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor, and shall be subjected to all provisions of this specification.

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3.3.1 Material deterioration prevention and control. The superstructure shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the superstructure may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the superstructure may be newly fabricated from recovered materials to the maximum extent practicable, provided the superstructure produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the superstructure.

3.3.3 Structural steel. Structural steel shall conform to ASTM A 36.

3.3.4 Low-alloy structural steel. Low-alloy steel shall conform to ASTM A 572/A 572M.

3.4 Welding electrodes. Welding electrodes shall conform to AWS A5.1 or A5.18, classification and size as required for the operation.

3.5 Flooring. Flooring shall be 5-inch I-beam, open steel floor, transverse-to-traffic type. Minimum section modulus shall be 1.68 inches cubed.

3.6 Bolts, nuts, and washers.

3.6.1 Bolts. Bolts shall conform to ANSI B18.2.1, hex head, steel, cadmium or zinc plated, UNC-2A.

3.6.2 Nuts. Nuts shall conform to ANSI B18.2.2, hex, plain, steel, cadmium or zinc plated, UNC-2B.

3.6.3 Washers. Washers shall conform to ANSI B18.22.1, round, steel, cadmium or zinc plated.

3.7 Connector pins. Connector pins shall be of steel conforming to ASTM A 304, UNS H41400, quenched-and-tempered and cold-finished to produce a minimum yield strength of 125,000 pounds per square inch (psi), minimum ultimate tensile strength of 140,000 psi, and a minimum elongation of 15 percent in 2 inches.

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3.8 Steel pipe. Steel pipe shall conform to ASTM A 53, black, standard weight, except that the ends shall not be threaded and the hydrostatic test will not be required.

3.9 Steel tubing. Steel tubing shall conform to ASTM A 513, MT 1020 steel, seamless, electric resistance welded or fusion welded type, as hot rolled or as welded condition, round shape.

3.10 Identification marking. Each component of the superstructure shall be metal-die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.11 Treatment and painting. All parts of each superstructure component except as specified herein shall be treated and painted in accordance with MIL-T-704, type B, except that the color chip number designation shall be 34088 in accordance with FED-STD-595. Pins, bolts, nuts, washers, and threaded portions of component when applicable shall not be painted but shall be preserved as specified in 5.1.

3.12 Stenciling. The nomenclature, Federal stock number, weight, and cube shall be stenciled on each major structural component of the superstructure such as the deck panels, filler panels, curbs, and ramp panels. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. The markings shall be Gothic-type capitals and Arabic numerals 3/4 inch high. Stenciling shall conform to MIL-STD-130.

3.13 Workmanship. The fabricated superstructure shall be free of sharp edges, slivers, or burrs.

3.13.1 Steel fabrication. Steel used in the fabrication of the superstructure shall be free from laminations, kinks, and sharp bends. Steel shall be cut by shearing, sawing, or flame cutting. Burned surfaces of flame-cut material shall be ground or machined to remove ash and cooling checks. The straightening of material shall be done by methods that will not cause injury to the metal. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.13.2 Welders and welding.

3.13.2.1 Welders. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code - Steel.
ASME Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

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Contractors who only make horizontal welds need not qualify welders for "all position welding". In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator. The test results shall be made available for review by the contracting officer or the contracting officer's representative.

3.13.2.2 Welding. The surfaces of parts to be welded shall be free from dirt, scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process. All welded parts shall be free from slag, flux, welded spatter, cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and adjoining base metal shall be wire brushed clean. Welding and welds shall be in accordance with either ASME or AWS code.

3.13.2.2.1 Welding jigs. Components parts to be welded shall be assembled in steel jigs or frames and welded while held in position. The jigs or frames shall be of a design that will limit the distortion to an amount that can be removed without changing the physical characteristics of the metal of the parts being welded.

3.13.3 Interchangeability. Like units shall be fabricated to be duplicates within the tolerances specified and shall be interchangeable.

3.13.4 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and on the applicable drawings. Tolerances for positioning of holes and parts shall be referenced from one end or side of the member and shall not be cumulative.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does

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not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for inspection shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when 0 defects are found and rejected when 1 or more defects are found.

4.4.2 Examination. Sample selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The superstructure shall be examined as specified herein for the following defects:

101. Dimensions not as specified (see 3.1).
102. Materials not as specified (see 3.3 thru 3.9).
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable operating and storage environments (see 3.3.1).
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
105. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.1.2).
106. Used, rebuilt or remanufactured components, pieces, and parts incorporated in the superstructure (see 3.3.2).
107. Identification marking not as specified (see 3.10 and 3.12).
108. Treatment and painting not as specified (see 3.11).
109. Workmanship not as specified (see 3.13).
110. Welding not as specified (see 3.13.2.2).

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

111. Unprotected surfaces not preserved as specified for level A.
112. Bolts, nuts, washers, and pins not packaged as specified and quantities exceeding the weight limitation of the box specification for level A.
113. Components bundled together in quantities other than as specified for level A or B.
114. Strapping not as specified for level A or B.
115. Shipping containers not as specified for level A or B.
116. Gross weight of wood boxes exceeds 200 lbs for level A.
117. Marking illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unprotected surfaces. Unprotected exterior metal surfaces of components such as pins, bolts, nuts, washers, and threaded portions of components, and any other unprotected metal surface requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.2 Bolts, nuts, washers, and pins. Bolts, nuts, washers, and pins of like description preserved as specified in 5.1.1.1 shall be packaged in close-fitting boxes conforming to PPP-B-636, W6c, in quantities not to exceed the weight limitation of the box specification. The boxes shall be waterproof sealed with tape as specified for slotted style boxes in accordance with the appendix to the box specification.

5.1.2 Commercial. The superstructure components, bolts, nuts, washers, and pins shall be preserved in accordance with ASTM D 3951 to afford protection against deterioration and damage during shipment from the contractor to the initial destination.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

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5.2.1 Level A.

5.2.1.1 Loose items. Panel assemblies; panels, deck, filler; panels, ramp, filler; deck units, short; or panels, ramp, tread shall be shipped loose.

5.2.1.2 Bundling. The structural components in the following quantities shall be bundled together: plate cover (4); plate, cover, filler (2); curb, complete, bridge (3); panel, deck, filler, short (3); curb, short, bridge (5); post, handrail and delineation marker (10); and beam, connector, bridge ramp (2). Strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc-coated, size as applicable, and ASTM D 4675.

5.2.1.3 Bolts, nuts, washers, and pins. Bolts, nuts, washers, and pins packaged as specified in 5.1.1.2 shall be packed together in close-fitting boxes conforming to PPP-B-621, class 2, style optional or PPP-B-601, overseas type, style optional, grade B. The gross weight of each box shall not exceed 200 pounds. The boxes shall be closed in accordance with the appendix to the box specification. Strapping shall be as specified in 5.2.1.2.

5.2.2 Level B. The superstructure components shall be packed as specified for level A except boxes shall conform to PPP-B-640, class 2, style as applicable, and strapping may be zinc-coated.

5.2.3 Commercial. The superstructure components as specified in 5.1 shall be packed in accordance with ASTM D 3951 to assure carrier acceptance and safe delivery to destination at lowest rates.

5.3 Marking.

5.3.1 Military. In addition to any special marking specified in the contract or purchase order (see 6.2), loose items, bundled items, and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Marking shall be in accordance with ASTM D 3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The superstructure assembly is intended for use as the deck section of the class 60 floating bridge.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required for inspection and approval, and the number of units required (see 3.2, 4.3 and 6.3).

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d. Degrees of preservation and packing required (see 5.1 and 5.2).
e. Any special marking required (see 5.3).

6.3 First article. When a first article inspection is required, the superstructure should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Bridge
Bridge superstructure
Curbs
Deck panels
Filler panels
Floating bridge
Pneumatic float bridge
Ramp panels
Steel superstructure
Superstructure

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME
Navy - MC

Preparing activity:
Army - ME

Project 5420-0189

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-117160	2. DOCUMENT DATE (YYMMDD) 930104

3. DOCUMENT TITLE
Superstructure, Steel, for Bridge, Floating, Pneumatic Float, Class 60

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Geraldine F. Harlow	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3469 (2) AUTOVON 654-3468
c. ADDRESS (Include Zip Code US Army Belvoir RDE Center ATTN: SATBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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